Video-Based Jump Height Estimation in Athletic Performance Testing

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***Abstract*—Jump height estimation is well-studied problem in sports science with a variety of methods proposed over the years. This paper introduces a portable, simple-to-setup and a cost-effective video-based jump height estimation method, and compares it with existing traditional methods that use expensive and specialized equipment. To enable more accurate comparison between different systems, a custom-designed Inertial Measurement Unit (IMU) was used as an additional sensor input. The analysis was performed on eleven female physically active handball players. The dataset consists of three most common vertical jump variations: squat jump, counter-movement jump and counter-movement jump with free arms. The analysis show that our video-based approach achieves comparable results to the traditional methods.**

***Index Terms*—jump height, testing, estimation, video analysis**

I. Introduction

Vertical jumping is considered to be an important factor amongst sport coaches and is used extensively for assessing athlete's potential, performance and overall ability [1]–[4]. Overall, vertical jump testing is an important tool for identifying areas for improvement in training and reducing the risk of injury [5]. Numerous methods exist for estimating an athlete's jump height and use different technologies [6], [7]. The results differ significantly because of the use of different approaches such as time of flight or force analysis. These methods may provide accurate measurements but are expensive because they require specialized equipment, and may not be feasible in all types of sports or training environments [8], [9]. In this analysis we considered three vertical jump variations: squat jump (SJ), counter-movement jump (CMJ) and counter-movement jump with free arms (CJMFA). The main difference between the SJ and CMJ jump variations is the involvement of a counter movement in the CMJ, which allows for greater potential energy storage and thus greater power output during the concentric phase. The SJ, on the other hand, relies more on the athlete's maximal strength and explosiveness, as there is no pre-stretch phase to generate additional force [9].

In recent years, video-based techniques have emerged as a more accessible and cost-effective approach for jump height estimation. In this paper we compare the performance of traditional methods with a video-based method. Our method is capable of high quality jump height estimation by analyzing video captured by low-cost cameras, such as present in mobile phones and other similar devices. The method utilizes a convolutional neural network to extract key features such as body landmarks to approximate the location of various body parts, from a video featuring a subject jumping. It uses those features to estimate the jump height [10]. The key body features are extracted using the MediaPipe Pose library. This library was chosen because of the easy-to use, low-code API. Other mainstream libraries [11] were not considered because of the lack of cross-compatibility between them.

II. Related work

Video analysis in software involves the use of computer algorithms to analyze and interpret the content of video footage [12]–[14]. This process can involve a wide range of machine learning techniques, including motion tracking, object and action recognition. The goal of video analysis is to extract useful information from video footage, such as identifying objects or people in the video, tracking their movements over time, or recognizing patterns or events that occur in the video.

The use of IMU devices for the purpose of estimating jump height is a fairly common practice. It has been shown that it can used as a simple and portable alternative to other methods [15]–[17]. Other benefits include the possibility of simultaneous monitoring multiple athletes. This approach is not covered by our video-based method and is something that should be taken into consideration. The comparison between force plates and optical measurement systems concluded that both systems give reliable and valid, but distinctly different results [6], [8], [18], [19].

III. Methods

The used video-based method works with a few assumptions. It assumes that the camera is positioned as such that the subject is visible during the entire duration of the jump. The video must be orientated as such that the jump is performed along the X or Y axis, as shown in Figure 2. Based on our testing, our method performs best when the camera is completely stationary and placed at subjects' hip height. The subject orientation does not play a significant role.

The input to the method is a video which contains a single subject performing a jump variation and the corresponding subject body height. The video should start with the subject fully upright, as shown in Figure 2, so the subject height can be used to approximate jump height. For accurate estimation, video should include the subject in the jump apex position. Since the method time of execution is linear with the given video length, it is recommended that the video is trimmed for best performance. Videos featuring multiple jumps do not result with multiple jump height estimates and the method should not be used as such. Although jump height estimation based on videos featuring multiple subjects in the background has during testing proved effective, the result is not guaranteed.

Time of flight can be easily extracted from the video by counting the exact number of frames it takes the subject from their feet leaving the ground to them reaching the apex. The jump duration in frames multiplied with the video frame-rate gives us the time of flight, as shown in Formula (1).

$$T_{flight} = \frac{N_{frames}}{Framerate} \tag{1}$$

Although this information was easily extracted using our method, we decided not to use it in the estimation because it showed to be too unstable. Unless video frame-rate is significantly greater than the mainstream 30 or 60 frames per second we do not recommend this method. If we consider a frame-rate of 30 frames per second, the time between two consecutive frames takes 1/30 seconds, what is roughly equal to 34 milliseconds. Since the jump height estimation based on time of flight is proportional to time squared and assuming an average jump height of 30-40 centimeters we conclude that the error of 1 frame could result in an error of $\sim$9 centimeters.

The center of mass is approximated at subject hip height by averaging the left and right hip features (represented as a 2d point), and the center of mass displacement in pixels can be easily extracted from the video by subtracting the vertical center of mass components of the start and the apex of the jump. Using video metadata (video height in pixels) and given subject height we can approximate how much a single pixel takes up vertical space in real life. With this information the jump height estimate can be easily calculated with mentioned information. Jump start is determined when subjects' feet leave the approximated five centimeter threshold area around the starting foot placement. This was done to reduce the error in jump height estimation when subjects feet moved slightly before takeoff. The jump apex is determined as the maximal height achieved by the approximated center of mass.

IV. Experimental setup

A. Participants

Eleven subjects participated in this experiment. Detailed information regarding the procedures, potential risks, and benefits of this experiment was clearly communicated to them before starting. The physically active subjects (Age = 24.18 $\pm$ 3.25 [years]; body height = 174.82 $\pm$ 7.61 [cm]; body mass = 68.69 $\pm$ 6.40 [kg]) were not engaged in any demanding exercises on the day of the experiment and they were instructed to perform to the best of their capabilities. Their personal data was collected prior to the experiment, with their written approval. The subjects performed three consecutive squat jumps (SJ), counter-movement jumps (CMJ) and counter-movement jumps with free arms (CJMFA) each. The jumps were completed in successive fashion with a brief period of recovery in between jumps.

B. Equipment

Jump height estimations is performed using the following devices: Quattro Jump (Kistler force plate), an optical measurement system OptoGait (Microgate), a smartphone camera which captures video at 60 frames per second and an IMU (Inertial Measurement Unit) device created by Ultrax Technologies. It was designed to be conveniently attached to subjects. Featuring its small size and its negligible weight, it is designed not to affect their jump ability. It is equipped with a 3-axis gyroscope, accelerometer and a magnetometer. The jump height estimation using the IMU device is estimated based on the recorded time of flight that is extracted from the recorded data.

The OptoGait panels are positioned on the very end of either side of the force plate. Using a tripod with a smartphone attachment, the camera is placed 3.5 meters away and 1 meter high. For reference, the upper edge of the platform was just shy of 13 centimeters. The experimental setup is shown at Figure 1.

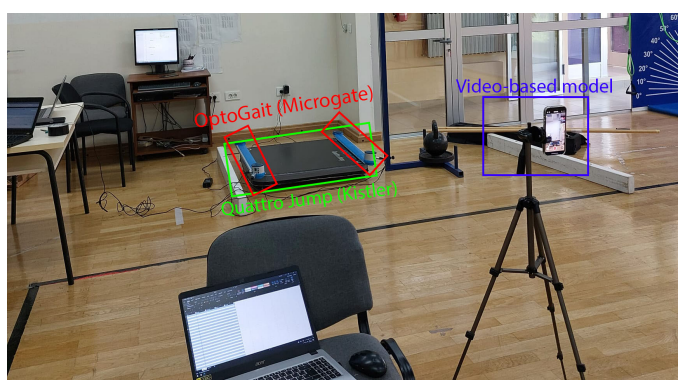

Fig. 1. Experimental setup

The subjects are facing away from the camera. At the start of the video, the subject is standing upright, in their starting positions for the corresponding jump variation. The camera is set up in such a way that the subject's feet are slightly above

the lower edge of the frame, ensuring that they remained fully visible throughout the duration of the video. The IMU device is placed on the back side of the subjects waistband using with the help of an integrated metal clip and adhesive tape to reduce IMU movement, which can be seen on figure 2. During our testing, the stability of the IMU device proved crucial for achieving valid results. The subjects relative positioning to the camera should not play a significant role, but for the purposes of testing it was useful that the IMU was also visible, to reduce human error.

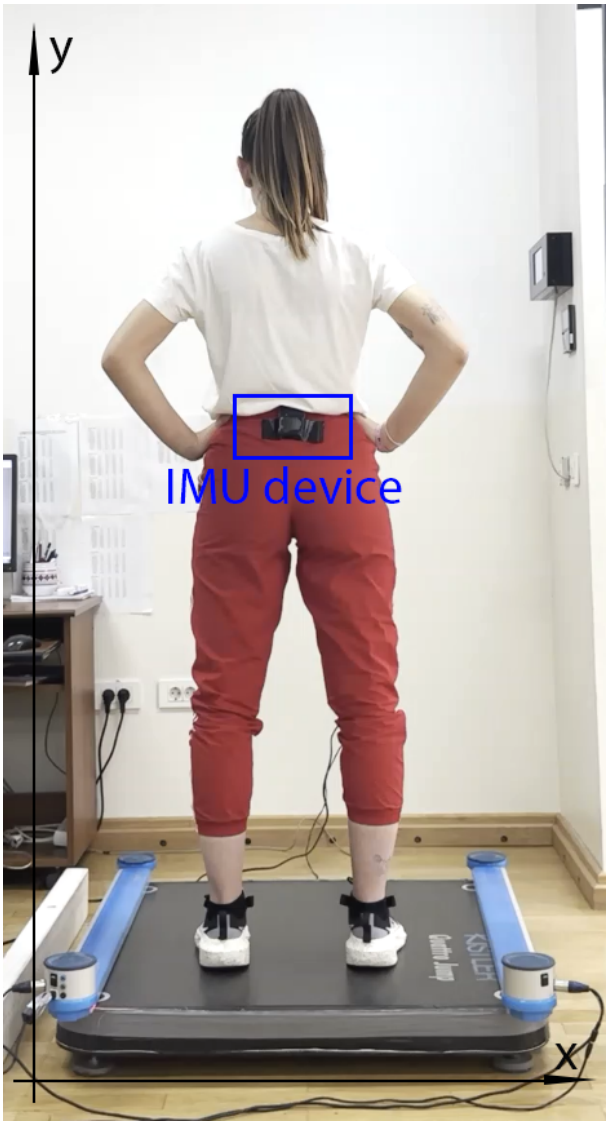

Fig. 2. Subject start position for SJ and CMJ

C. Procedure

Prior to the test protocol execution, all of the equipment was turned on, calibrated and prepared for the uninterrupted recording. The preparation consisted of inputting the subjects' data in the Kistler force plate and OptoGait software, such as their full name, height, mass and date of birth. Test jumps were performed by one of the experimenters to verify that every part of the setup is functional and ready to use. The eleven subjects were separated into three (almost) equally sized groups and each group was pre-assigned a time slot of 45 minutes. The group size was determined as such that the jumps could be comfortably completed within the assigned time slot. This approach optimized the recording efficiency without compromising the overall effectiveness. Every group completed all of their jumps (9 consecutive jumps that consisted of 3 attempts for each jump type, with a short rest period in between jumps) before the start of the next group. For each group, the subjects then performed the jumps which were simultaneously recorded and stored for further analysis and comparison. The created dataset was exported and analyzed at a later date.

V. Results

The coefficient of variation, show in formula (2)

$$CV = \frac{\sigma}{\mu} \tag{2}$$

is a statistical measure commonly used to assess the relative variability or consistency within a dataset. In this study, the CV was employed to evaluate the consistency of jump height measurements across the different types of jumps for each measurement method. The results indicated that the method utilizing video analysis, exhibited the most consistent measurements across all three jump types, with CV values ranging from 1% to 9%, followed by OptoGait with CV values ranging from 1% to 15%. For reference, our IMU device also performed very consistently (especially on SJ) with a CV of around 10%.

The Kruskal-Wallis test is a statistical tool used to compare the medians of three or more independent groups, and it does not require normality or equal variances in data. This test is often used as an alternative to ANOVA, which is a parametric test that assumes normality and equal variances. The Kruskal-Wallis test is especially useful when analysing skewed or non-normally distributed data. This study utilizes the Kruskal-Wallis test because the collected data was not normally distributed and showed unequal variances among the five independent groups. The Kruskal-Wallis test revealed a significant statistical difference among the group medians ($H = 262.78$, $p < 0.001$). This means that the null hypothesis, which assumes equal medians among the groups, can be rejected. The observed differences in medians are therefore unlikely to be due to random chance alone.

The Bland-Altman test is a widely used statistical tool for assessing the agreement between two measurement methods or comparing a new method against a reference method [21], [22]. It provides valuable insights into the average difference between the measurements and the level of agreement or disagreement between them. In our research, we employed this method to evaluate the agreement of all five measurement methods. For each combination of measurement methods, we conducted a Bland-Altman analysis and plotted the results, generating scatter plots that provided insights into

the disagreement patterns among the methods. Each scatter plot included three subplots, one for each jump type under investigation. Analysing the scatter plot results on Figure 3, we observed that the combination of the video analysis and Kistler device demonstrated the highest agreement level across all three jump types, with minimal bias and relatively narrow limits of agreement. The video analysis and OptoGait device, along with the video analysis and the IMU device pairs also exhibited favourable agreement as shown in Figure 4 and Figure 5, respectively. Since the continuous processing of the IMU orientation is still in development, we notice the discrepancy in the data. To present our findings more concisely, we included Bland-Altman plots for the best-performing pairs. The figure offers a clear overview of the agreement between these pairs, highlighting their suitability for precise measurement of jump heights.

Fig. 3. Agreement of Jump Heights: Video and Kistler

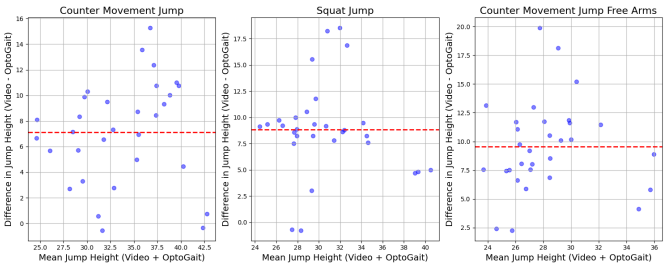

Fig. 4. Agreement of Jump Heights: Video and OptoGait

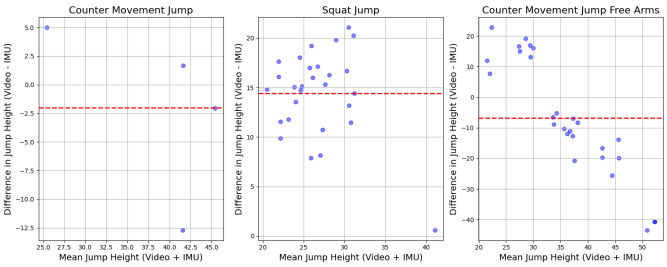

Fig. 5. Agreement of Jump Heights: Video and IMU

VI. Discussion

The small sample size and the similarities in jump height estimate may be the limiting factors in the analysis. Human error and setup may play a significant role during testing [23].

It is worth mentioning that video-based analysis has its limitations. Although we did not use the time of flight approach, we recognize that the accuracy of the method is significantly compromised when comparing the capability of a standard smartphone, with its typical video frame-rate of 60 frames per second, to the polling rate of specialized equipment that can reach upwards of 1000 hertz or more. The main drawback is that the time-of-flight estimate granularity is significantly reduced. The relative positioning and the angle of the camera can also play a significant role in the accuracy of the method. If the camera is not positioned properly or at the right angle, it can lead to inaccurate results.

Several commercial products utilize video technology to estimate jump height, including My Jump Lab [24]. However, these products typically require the user to identify the precise frame when the subject takes off and lands, which we chose to avoid in our approach and in our testing. This decision was made because we believe that this approach is cumbersome enough for the user and that the benefits of users manual input do not outweigh the drawbacks.

It is also important to take into account the time of estimation of our method. Unlike the practically instantaneous results provided by Kistler and OptoGait, our method's time of execution is linear with the duration of the video. For reference, force plates typically use the double integration method [20] which requires less computational resources compared to video-based methods. This method is also subject to higher degrees of human error compared to other methods [12]. Although our method supports varying levels of complexity, there is no significant difference in its performance despite significant variations in the time required for execution. Specifically, In terms of method complexities (basic, medium, complex), the average execution times for a 3-second video were 3 seconds, 8 seconds, and 15 seconds, respectively.

Finally, having a low-cost and easily accessible method for measuring jump height is important for enabling easier access to sports science research [16]. It can facilitate a wider range of researchers to investigate the relationship between jump height and other factors such as genetics, training, and injury risk, ultimately leading to new insights and advancements in sports science.

VII. Conclusion

The goal of our approach was to provide a simple and accessible method for jump height estimation. The method utilizes a convolutional neural network to extract key features such as body landmarks which approximate the location of various body parts, from a video featuring a subject jumping and uses those features in the jump height estimation. The results are comparable with traditional methods. Additionally, our video based method has shown to be the most consistent amongst all of the tested methods. Taking into consideration that the jumps were consistent, this can be considered as a indication of good performance. Since this testing method is more cost-effective and easier to implement it has the potential to be a valuable tool for coaches, trainers and athletes to

monitor and improve their jumping ability. The widespread availability of the smartphone, along with the portability of software makes it a viable solution for amateurs, enthusiasts and professionals in a wide variety of complex environments. More research is required because there is a lot of possible room for improvement, and future studies should consider experimenting with different jump types, camera positions and other models.

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